

Ground(less) Truth: The Problem with Proxy Labels in Human-AI Decision-Making

Working Draft. Comments Welcome.

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ABSTRACT

A growing literature on human-AI decision-making investigates strategies for combining human judgment with statistical models to improve the quality of decisions. Existing research in this area often evaluates proposed improvements to models, interfaces, or workflows by demonstrating improved predictive performance on “ground truth” labels. However, this practice overlooks a key difference between human judgments and model predictions. Whereas humans commonly reason about broader phenomena of interest in a decision – including latent constructs that are not directly observable, such as disease status, the “toxicity” of online comments, or future “job performance” – predictive models target *proxy* labels that are readily available in existing datasets. Predictive models’ reliance on simplistic proxies for these nuanced phenomena makes them vulnerable to various sources of statistical bias. In particular, we identify four sources of *target variable bias* that can impact the validity of proxy labels in human-AI decision-making tasks. We develop a causal diagram that disentangles the relationship between each bias and clarifies which are of concern in specific human-AI decision-making tasks. We demonstrate how our framework can be used to articulate implicit assumptions made in prior modeling work, and we recommend evaluation strategies for verifying whether these assumptions hold in practice. We then leverage our framework to re-examine the experimental design of prior human subjects studies investigating human-AI decision-making, finding that only 6 of 72 reviewed studies examine factors related to target variable bias. We conclude by discussing opportunities to better address target variable bias in future human-AI decision-making research.

1 INTRODUCTION

A growing body of research combines predictive machine learning models with human judgment to improve decision-making processes. This work often pursues the goal of *complementary performance*: configurations of humans and models that yield higher-quality decisions than either would make in isolation. In the machine learning community, model-level improvements have been proposed to address gaps in human judgment (e.g., [52, 69, 102, 108]). In the human-computer interaction community, behavioral interventions have been developed to help *humans* better incorporate model outputs into their decision-making (e.g., [5, 12]).

However, current evaluations of complementary performance make the key assumption that the label targeted by a predictive model adequately reflects the goals of human decision-makers. Specifically, decision quality is frequently operationalized by predictive performance, measured via accuracy, AU-ROC, or similar statistical measures computed with respect to “*ground truth*” or “*gold standard*” labels recorded in available data. The relative performance of human judgment, model predictions, and hybrid combinations of the two are then ranked according to these metrics. Yet these notions of decision quality hinge on the definition of “*ground truth*” considered by the model. Consequently, comparisons of human and model performance are only valid insofar as the “*ground truth*” label targeted by the model reflects the underlying objectives of human decision-makers.

Yet in real-world human-AI decision-making settings, labels are often imperfect proxies for the target outcomes considered by decision-makers. While making decisions, humans frequently consider latent constructs such as the “toxicity” of an online comment, “cardiovascular disease risk” of a patient, or future “job performance” of a candidate. Because observed labels (e.g., toxicity annotations, diagnostic test results, and performance reviews) serve as indirect measurements of these phenomena [55], they can be subject to *measurement error*. Additionally, humans often select among multiple possible actions (e.g., medical treatments, social welfare interventions) in hopes of improving a downstream outcome of interest. Because an outcome is only observed for the selected action, labeled data does not contain the counterfactual outcome that *would* occur had a different option been chosen instead. This introduces a set of additional challenges, including selective labels [66], intervention effects [98], and selection bias, which interact with measurement error in nuanced ways depending on the nature of the decision-support task. In this work, we refer to this collection of challenges – which can be characterized as sources of statistical bias impacting labels – as *target variable bias*.¹

Target variable bias has been widely documented in real-world deployments of algorithmic systems (e.g., Table 1). For instance, predictive models impacted by target variable bias have contributed to

¹The term target variable bias was initially introduced by Fogliato et al. [38] to describe a specific form of measurement error. We use this as an umbrella term that can also encompass other sources of statistical bias impacting the target variable.

unwarranted firing of teachers [18], perpetuated historical disparities in access to medical resources [81], and raised concerns among social workers investigating allegations of child abuse and neglect [20, 57]. Surprisingly, existing modeling efforts and human subjects experiments in the human-AI decision-making literature have largely overlooked this challenge. Left unaddressed, this disconnect could undermine the ultimate goal of human-AI decision-making research: to develop algorithmic systems that meaningfully improve decision-making in real-world contexts. In this paper, we aim to bridge the gap between challenges encountered in real-world deployments of predictive models and current human-AI decision-making research practices by raising awareness of target variable bias, identifying blind spots in previously published modeling approaches and human subjects experiments, and providing guidelines for improved model evaluation and experiment design practices going forward.

Based on an examination of numerous real-world ADS deployments, we construct a *causal diagram* representing the data generating process in ADS. This diagram (1) distills high-level structural differences between decision support tasks (e.g., content moderation, recidivism prediction, and child welfare hotline screening, among others) and (2) disentangles which sources of bias are of concern in each setting. Informed by discussions with researchers across several disciplines, we map our causal diagram to existing terminology and methods used to characterize target variable modeling assumptions in learning sciences, biostatistics, psychology, and economics.

We use our framework to identify strategies for better-addressing target variable bias through two lines of human-AI decision-making research:

- **Model development.** We develop a two-step *measurement* and *prediction* decomposition for specifying target variable modeling assumptions. We use this decomposition to create a taxonomy of previous model-level ADS improvements proposed in the literature and identify a recommended set of measurement model evaluation strategies.
- **Experimental human subjects studies.** We re-examine the design of prior human-AI decision-making experiments through the lens of our causal diagram. Through our analysis, we identify systematic blind spots in our current understanding of algorithm assisted *human* decision-making in settings where predictive models are impacted by target variable bias.

2 RELATED WORK

In this section, we first overview a body of research on human-AI decision-making, which our framework is intended to inform. We then summarize modeling challenges and broader validity concerns that draw into question current research practices (i.e., modeling assumptions, experimental study designs, and measures of decision quality) adopted in human-AI decision-making literature.

2.1 Human-AI decision-making

A body of research in machine learning has developed *modeling approaches* with the explicit aim of complementing the limitations of human judgement. Drawing from a long line of work showing that actuarial risk assessments can outperform expert judgement in

many prediction tasks [29, 46], recent methods have been proposed that learn to complement humans by adaptively routing decision instances [42, 69], leveraging heterogeneity in human and machine decision performance [19, 36, 102, 108], leveraging consistency in expert decisions [31], and adapting to [52] and training [75] human mental representations of model outputs. Yet such modeling approaches operate on a set of *simplifying assumptions* about the world, which may or may not hold in a given deployment context. We provide a framework for articulating modeling assumptions, and show that many common assumptions made by prior work involving proxy labels are unlikely to hold in practice.

In addition to model-level improvements, growing research has studied opportunities for human-AI complementary in algorithm-assisted *human* decision-making [12, 16, 41, 44, 64, 65]. This work investigates the potential for tools such as training protocols [15, 16, 64], explanations [13, 67], and other behavioral interventions [12, 41], to improve how humans make use of model outputs. While many online experimental studies have been focused on interventions to improve predictive performance, little work to date has experimentally studied other key factors that are present in real-world deployment contexts, such as asymmetric access to information [51, 53], measurement error [43], and omitted payoffs [45].

2.2 Modeling challenges in algorithmic decision support

Previous work has surfaced a litany of challenges impacting predictive models in decision support tasks. Kleinberg et al. [59] examine the impact of *unobservables* – or predictive attributes that are available to humans, but are not available for modeling (e.g., because they are not easy to collect or encode in datasets). Lakkaraju et al. [66] consider the problem of *selective labels*, which occur when an outcome is only observed under one of the possible decisions (e.g., when default outcomes are only observed among lenders granted a loan). Selective labels are of concern in the common setting where past decisions were assigned non-randomly [31, 59, 66]. Here, *selection bias* can cause the distribution for which outcomes are observed (e.g., lenders approved for a loan) to differ systematically from the general population (e.g., *all individuals*, including those at high risk of loan default) [30, 99].

A related modeling challenge arises when decisions serve as *interventions* intended to mitigate risk [25]. Though it is often desirable to estimate the likelihood of a target outcome (e.g., heart attack) under multiple actions (e.g., treatment options), an outcome is only observed for the decision selected in the past. While counterfactual modeling approaches have been developed to predict the outcome that *would occur* under different hypothetical actions, these techniques are only reliable in settings with limited unobservables (i.e., unmeasured confounding [23, 86]) and selection bias (i.e., positivity [79, 86]).

Additional work has examined the quality of proxy labels in decision support tasks. Obermeyer et al. [81] studied “*label choice bias*”, in which racial disparities in access to health resources were introduced by poor proxy selection decisions. “*Omitted payoffs bias*” has been used to describe factors of interest to humans that are incompletely reflected by predictive models targeting available labels [18, 31, 59]. While this bias describes challenges specific to

Work	Domain	Form of evidence	Bias reported
Kleinberg et al. [58]	Judicial	Empirical analysis	Examines role of unobservables, selection bias, and proxy measurement error in judicial decision-making. Introduces term of omitted payoff bias
Fogliato et al. [40]		Empirical analysis	Examines measurement error introduced by adopting re-arrest as a proxy for re-offense
Bao et al. [6]		Synthesis of RAI and criminology literature	Discusses selection bias and measurement error challenges introduced by recidivism RAIs
Butcher et al. [14]		Criminology literature	Reports group-dependent measurement error arising from differential arrest rates among Black and white drug crime offenders
Kawakami et al. [57]	Child Welfare	Contextual inquiry, interviews	Documents social worker concerns that measurement error and unobservables impact the quality of ADS predictions
Cheng and Stapleton et al. [20]		Empirical analysis, contextual inquiry, interviews	Documents social worker concerns that measurement error impacts the quality of ADS predictions
Andrus and Gilbert [2]		Methodological framework	Investigates ADS tools as a method of surfacing measurement error, construct validity, and fairness concerns in institutional practices
Obermeyer et al. [81]	Medical	Empirical analysis	Reports group-dependent measurement error arising from a <i>cost of care</i> health proxy used for medical need
Mullainathan and Obermeyer [76]			Characterizes measurement error introduced by adopting recorded stroke as a proxy for biological stroke
Mullainathan and Obermeyer [77]			Documents implications of unobservables, selection bias, and measurement error in heart attack testing decisions
Chalfin et al. [18]	Hiring	Empirical analysis	Discusses omitted payoffs, measurement error, and selection bias arising in <i>teacher value-add</i> proxy used for educator hiring

Table 1: Documented examples of target variable bias impacting predictive models across numerous ADS domains.

prediction (e.g., model unobservables, measurement error [58]), this term also applies when humans care about a broader set of decision-making factors beyond risk of a single adverse outcome [18, 44]. In this work, we primarily use the framework of *measurement error* and *validity* while discussing systematic differences between target outcomes of interest to humans and proxy labels observed in data [55]. This framework is particularly useful because it draws upon a rich set of existing knowledge and methodologies previously developed in adjacent disciplines (e.g., psychology, political science, sociology) for examining how latent phenomena of interest to humans are quantified in observed data [94].

2.3 Measurement and validity in algorithmic systems

Recent work in machine learning and algorithmic fairness has raised broader concerns regarding whether algorithmic systems successfully achieve their purported function. Synthesizing concepts from measurement theory in the quantitative social sciences, Jacobs and Wallach [55] argue that unobserved latent constructs of interest to humans (e.g., creditworthiness, recidivism, and fairness) are frequently *operationalized* via *measurement models* in algorithmic systems. Further, these operationalizations of latent constructs (e.g., loan default outcomes, re-arrest outcomes, and statistical fairness metrics) can be impacted by *measurement error* when they imperfectly reflect the construct they are intended to

measure [48, 94]. Jacobs and Wallach [55] describe components of construct reliability and validity that can be used to assess the quality of an operationalization of a construct during evaluation of algorithmic systems. Lundberg et al. [68] develop a three-step framework for reifying the relationship between theoretical and empirical estimands of interest in sociology research. This approach is similar to our two-step framework distinguishing between measurement and prediction in predictive modeling (Section 4).

Recent work has highlighted validity concerns specific to algorithmic Risk Assessment Instruments (RAIs). Bao et al. [6] examine statistical biases present in criminal justice datasets (e.g., ProPublica’s COMPAS Dataset [4]) used for algorithmic fairness benchmarks. This analysis identified several biases in the *outcome variable* Y predicted by models, which we further characterize in this work. Coston et al. [22] examines a host of validity concerns impacting RAIs, including many discussed in Section 2.2. This work proposes an interview-based protocol for navigating validity concerns among multiple stakeholders when deciding whether deployment of an RAI is justified. Recent work has also surfaced validity concerns in settings such as content moderation [43] and recommender systems [74, 101].

Despite growing awareness of the importance of measurement and validity in algorithmic systems, we currently lack a holistic understanding of validity threats to prediction targets in human-AI decision-making. Such an understanding is critical for developing reliable measures of human-AI decision quality and preventing

algorithmic harms in real-world deployment contexts (e.g., Table 1). Therefore, in this work, we use causal diagrams to examine the relationship between measurement error and other modeling challenges (e.g., unobservables, selection bias, and intervention effects) that can impact the validity of prediction targets in real-world decision support settings. To our knowledge, our work offers the first holistic discussion of how measurement error, unobservables, selection bias, and treatment effects interact to impact target variable validity in real-world ADS deployments.

2.4 Causal diagrams

We propose a *causal diagram* that models the data generating process in real-world ADS scenarios. We use our causal diagram to (1) clarify the relationships between different statistical biases that can impact prediction targets and (2) articulate which sources of bias are of concern in specific human-AI decision-making tasks. Causal diagrams are Directed Acyclic Graphs (DAGs) that show causal relationships between nodes using directed edges [85]. While structural causal models (SEMs) encode additional statistical details regarding *how* multiple variables are related [11], causal diagrams provide a higher-level description of *whether* two variables are related via conditional independence statements [86]. We develop a causal diagram in favor of an SCM because our analysis of target variables is intended to identify biases that occur based on *whether* different variables are correlated while generalizing over the specifics of these statistical relationships, which differ across domains.

Our use of causal diagrams to examine proxy labels parallels existing work using causal modeling for algorithmic fairness [8]. In algorithmic fairness contexts, causal modeling has been used to assess whether observed outcome disproportionalities constitute fairness-related harms [28, 70]. Because causal diagrams are readily interpretable by both technical and non-technical stakeholders, they are well-equipped to facilitate such deliberations [8, 17].

3 FRAMEWORK

We now describe our framework scope (Section 3.1) and development process (Section 3.2) before introducing the causal diagram upon which our framework is based (Section 3.3). We then use our framework to disentangle various sources of target variable bias (Section 3.4).

3.1 Scope

Our framework is designed to examine target variable bias in human-AI decision-making settings where a statistical machine learning model is introduced to augment human judgement. In particular, we examine settings in which a supervised learning model is trained to predict a label such as (i) a future event (e.g., medical [81], criminal justice [35], child welfare [21], or real estate [53, 91] related outcomes); (ii) a subjective human annotation (e.g., perceived content toxicity [43]); or (iii) factual information (e.g., food nutrition [12]). In these settings, model predictions are combined with human decision-making, either by showing model predictions to a human (i.e., algorithm-in-the-loop [44]), who makes the final decision, or via a more nuanced hybrid flow of autonomy (e.g., deferral-based learning [69], learning with bandit feedback [42], or conditional delegation [62]). This *algorithmic decision support* (ADS)

setting has received substantial research interest given increasing real-world prevalence of related systems.

The terms “human-AI decision-making”, “human-AI collaboration”, and “human-AI complementarity” have recently been used to describe a range of collaborative tasks involving humans and statistical models. We use the term “predictive model” rather than “AI system” or “AI-based decision-support tool” to disambiguate settings outside the scope of this work. Given our focus on prediction-based decision-making tasks, we do not examine settings where unsupervised learning (e.g., clustering) is used to surface patterns in data to support decision-making. Further, our framework does not address creative tasks involving *generative models*, such as image generation. Finally, while our framework may be useful in sequential settings in which the target variable is time-dependent (e.g., reinforcement learning, contextual bandits), we do not directly consider such settings in this work.

3.2 Framework development

Prior studies of real-world ADS deployments have surfaced numerous examples of target variable bias impacting predictive models [6, 18, 40, 57, 58, 76, 81]. Yet, these challenges have been acknowledged to varying degrees in prior modeling work (i.e., Table 2) and they are rarely examined in human subjects experiments on human-AI decision-making (i.e., Figure 3). A key barrier to better understanding and addressing these challenges is that we currently lack a holistic view of the statistical biases that can impact target variables. A second barrier arises because different ADS tasks (e.g., content moderation, forest cover prediction, heart attack prediction) are subject to distinct sets of challenges. Tasks involving the prediction of factual information (e.g., nutrient content of food [12]) introduce a different set of challenges than those involving the prediction of subjective or socially contested outcomes (e.g., child welfare outcomes [21]). Understanding which statistical biases are of concern in a given ADS task requires examining the historical *data generating process* that produced the dataset used to train a predictive model.

Causal diagrams describe causal relationships among variables in a system, and are often visualized via Directed Acyclic Graphs (DAGs), which intuitively communicate the flow of causal information across variables. A DAG shows each variable being modeled as a discrete *node*. Causal connections between variables are shown as *directed edges* connecting nodes, where each edge represents a causal path from the parent to its child. Sometimes, we assume the state of a variable is known in advance, but do not attempt to model its causes. These *exogenous* variables are shown in DAGs by nodes with no incoming edges. In other cases, we explicitly model how a variable will respond to changes to its inputs. These *endogenous* variables are shown in DAGs as nodes with inbound edges [86].

Our framework examines challenges impacting the labels available in data. Therefore, in our causal diagram, we specifically consider variables (i.e., *nodes*) and relationships (i.e., *edges*) that directly relate to the target variable. As such, our diagram *abstracts away* other key factors that can vary across contexts, such as the training [15, 16, 64], decision-making process [45], and workflow [44] of the human experts using the predictive model. While prior work has examined these factors in detail [15, 16, 44, 45, 64], our diagram

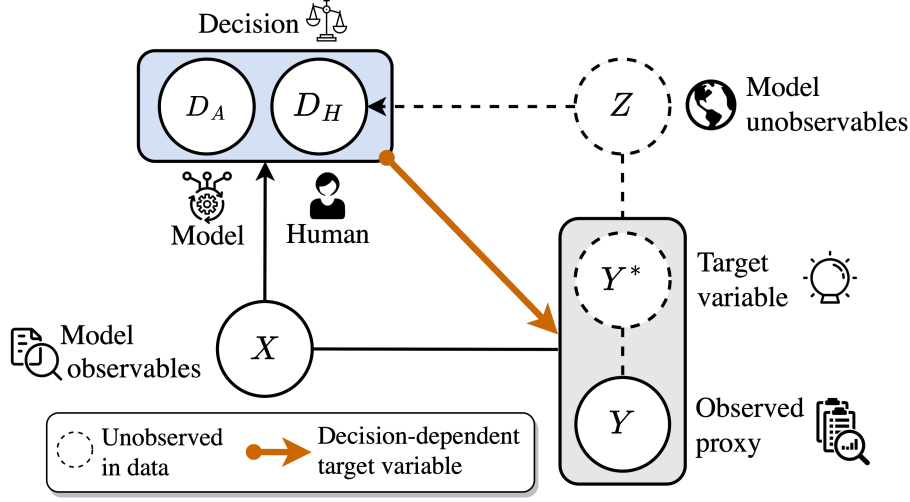


Figure 1: Our proposed causal diagram. X describes recorded information that is observable to a model via covariates. In some cases, a subset of this information may be unavailable to humans. Z describes contextual information that is available to humans but remains unobserved as model covariates. The blue shaded box shows decisions, which may result from (1) human-only decisions D_H pre-deployment of the model or (2) a function of human decisions D_H and algorithmic decisions D_A post-deployment of the model. The grey shaded box shows the prediction target, including the target variable of interest to humans (Y^*) and its observed proxy (Y). Observed variables are shown with solid lines, while unobserved variables are shown in dotted lines. An arrow pointing to a shaded box is shorthand for separate arrows pointing from the source to each of the nodes contained within the box. Edges with directionality that varies across ADS tasks are indicated via bidirectional edges.

foregrounds factors most salient for understanding target variable bias. In Section 5.3, we outline how our causal diagram can be extended to systematically examine a broader set of components beyond target variables in ADS research.

Our causal diagram was developed and refined through an iterative series of discussions among the authors and external researchers spanning a range of disciplines. Based on a review of real-world case studies (i.e., Table 1), we synthesized candidate models that could adequately characterize the target variable of interest across settings, and then stress-tested these models by attempting to identify counterexamples. Through our discussions with external researchers, we also cross-referenced our framework with existing terminology and methods developed in adjacent disciplines, such as medical diagnostic testing, educational assessment, mental health, and statistics.

3.3 Causal diagram

3.3.1 Diagram structure. We begin by describing the variables included in our causal diagram, which are shown as nodes in Figure 1. X describes observed attributes available as model covariates. In some cases, human decision-makers can be unaware of a subset of covariates (e.g., due to organizational policy, legal restrictions, or prohibitively large datasets) [53]. Z describes predictive contextual information that is available to human decision-makers, but is not recorded in a form suitable for modeling. Figure 1 refers to X and Z as *model observables* and *model unobservables*, respectively, based on whether the information is available to the model. Y^* describes the unobserved target outcome of interest to human decision-makers, while Y represents its proxy recorded as a label in data.

Given that models are commonly introduced to augment existing decision-making processes, past decisions D are often available during model development. Prior to the deployment of ADS, decisions typically result solely from human judgment D_H . After a model has been deployed, decisions can result from a combination of human judgement D_H and algorithm predictions D_A . We show this joint decision combining D_H and D_A in the blue shaded box of Figure 1. In some cases, joint decisions result from humans incorporating algorithmic recommendations in their decision-making (i.e., algorithm-in-the-loop [44]). In other cases, the joint decisions result from a learned combination of D_H and D_A [19, 36, 42, 52, 69, 102, 108].

We now consider causal relationships between nodes in Figure 1. First, we describe all of the possible relationships that *can* hold across ADS tasks, before narrowing in on special cases that carry implications for target variable bias in Section 5.1. Across ADS tasks, observed covariates X and unobservables Z both contribute to human decisions D_H , while algorithmic predictions D_A are only influenced by covariates X . We show these relationships via directed arrows from X to D and Z to D_H . Decisions D can also influence the target outcome Y^* and/or the proxy label Y (i.e., as when medical treatments impact health outcomes and diagnoses). We show this relationship via an arrow from D to Y and Y^* .

Covariates X and model unobservables Z are also related to target variables Y and Y^* . However, the directionality and specific causal pathway (including additional mediators or confounders) varies depending on the specific ADS domain. We show this variation in directionality via undirected edges between X and Y/Y^* as well as Z and Y/Y^* . We elaborate on examples and implications of this nuance in Section 3.3.2. Finally, the target variable of interest

Y^* is *operationalized* by a proxy Y via a measurement model. We show a bidirectional edge between Y and Y^* to indicate that the directionality of causation between Y and Y^* can also vary across ADS domains. All edges shown with bidirectional arrows simplify to a single directed edge in a specific ADS modeling context.

3.3.2 Task-specific graph structure. A key benefit of our causal diagram is that it enables identifying structural similarities and differences between ADS modeling tasks. While Figure 1 presents the set of all possible edges, some of these connections do not apply in specific ADS settings. By mapping ADS modeling tasks to special cases of our causal diagram, we can identify and mitigate sources of target variable bias that are applicable in each setting.

One factor distinguishing among tasks is whether the target variable depends on the decision being informed by a model. Settings with *decision-dependent target variables* occur when the prediction target of interest is influenced by the proposed decision (see orange arrow in Figure 1). Prediction of future medical, financial, social welfare, or education outcomes typically involves decision-dependent target variables. This is because decisions in these settings constitute *interventions* (e.g., medical treatments, educational programs) or *opportunities* (e.g., loans, new candidate hires, release on bail) that change the likelihood of the target outcome (e.g., loan default, disease prognosis, educational attainment).

Conversely, the target variable is *not* influenced by the proposed decision in *decision-independent* modeling tasks. While tasks with decision-independent target variables are uncommon in real world deployments of algorithmic systems, these tasks are common in experimental human-AI decision-making studies. For instance, studies have examined models that predict factual content (e.g., food nutrition [12]) and perceptual information (e.g., counts of objects [83], geometric shapes [109]). These tasks involve decision-independent target variables because the prediction target (i.e., food nutrition, object quantity, geometric shape) is not influenced by the prediction made by a human and/or model. Because decision-independent modeling tasks introduce fewer sources of target variable bias (Section 3.4), this carries implications for the ecological validity of experimental studies (Section 5.2).

An additional special case of our causal diagram involves tasks in which the target variable is a latent construct (e.g., toxicity, hate speech, or emotional state) defined by *subjective human annotations*. In these settings, human annotations serve as a *measurement model* linking the target construct (e.g., toxicity, hate speech) to a proxy (i.e., ratings from a specific set of individuals). Because humans can disagree considerably on subjective tasks, labels elicited from a specific set of individuals (Y) serve as an imperfect proxy for the broader construct of interest (Y^*). In subjective human annotation tasks, D_H and D_A constitute decisions made by a joint human-model system at *runtime* (i.e., after deployment of system predicting the target construct). For example, these decisions can involve removal of content flagged as “toxic” or “hate speech” [61], and reversal of Wikipedia edits deemed to be “damaging” [47].

Figure 1 also highlights domain-specific variation in directionality of causation between covariates X and a prediction targets Y/Y^* . Tasks such as predicting loan default, defendant recidivism, or health outcomes are most commonly framed as forecasting future observed outcomes Y given past behavior represented by observed

covariates X and unobservables Z . For this reason, causal diagrams in algorithmic fairness and predictive modeling commonly show a directed arrow $X \rightarrow Y$ [9, 49]. However, in some settings, as when a latent disease state Y^* manifests via records with pathological physiology X , the flow of causal information can run in the opposite direction such that $Y^* \rightarrow X$. This same relationship can also occur with unobservables such that $Y^* \rightarrow Z$.

This variation in edge directionality can introduce considerable nuance to predictive modeling tasks. For example, it is conceivable that disease status (Y^*) could impact observed symptoms (X), which then impacts downstream treatment (D) and observed case outcomes (Y). In order for the causal diagram to remain valid (i.e., a directed *acyclic* graph), one of the edges connecting nodes must remain disconnected in these settings (i.e., when target variables are decision-dependent and $Y^* \rightarrow X$, $Y \rightarrow X$, $Y^* \rightarrow Z$, or $Y \rightarrow Z$). While this requirement is consistent with the scope of our framework, which considers non-sequential settings, feedback loops are an important factor to consider in sequential settings [37]. Yet, the sources of target variable bias we highlight in the following section occur even in straightforward non-sequential settings.

3.4 Sources of target variable bias

We now use our causal diagram (Figure 1) to examine four sources of target variable bias that can threaten the validity of target variables in ADS tasks. Some biases (e.g., measurement error) can occur in all ADS tasks, while others (e.g., intervention effects, selection bias, and confounding) are of concern in tasks with decision-dependent target variables.

3.4.1 Measurement error. Human experts often make decisions involving unobserved, latent constructs such as “cardiovascular disease risk”, “student learning”, and “job performance.” These latent constructs are not directly observable in the world, but can be operationalized via a *measurement model* [48, 55]. Adopting a label observed in historical data as a proxy for an unobserved latent construct serves as a *de facto* measurement model. For instance, in criminal justice settings, defendant re-arrest is commonly adopted as a proxy for recidivism [6, 40], while in commercial hiring settings, manager reviews are frequently adopted as a proxy for future job performance.

Measurement error (Figure 2.A) occurs when there is a systematic difference between the target variable of interest to human decision-makers (Y^*) and its operationalization by a proxy (Y). One notable case of measurement error occurred in the medical domain, where a widely-deployed clinical decision support tool intended to inform physician decision-making used “*cost of medical care*” as a proxy for patient medical need [81]. However, due to systemic disparities in healthcare access, Black patients historically received less care (and thus had lower medical costs) than similarly sick white patients. As a consequence, these patients were turned away from much-needed medical resources. This challenge has also been extensively documented in other ADS deployments in judicial [14, 38], child welfare [20, 57], and hiring [18] domains (i.e., Table 1).

Critically, proxy labels impacted by measurement error are an incomplete reflection of the actual goals of human decision-makers, and therefore serve as an incomplete measure of human-AI decision quality. Therefore, before adopting a proxy as a measure of decision

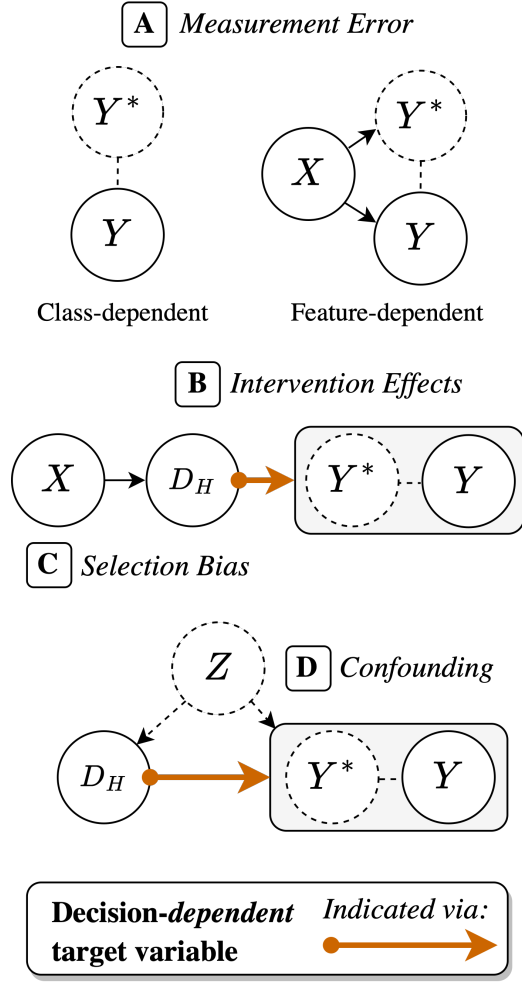


Figure 2: Sub-graphs of the DAG in Figure 1 introducing statistical biases that impact the target variable Y^* . *Measurement error* (A) can occur in settings with both decision-dependent and independent target variables. In decision-dependent settings, *intervention effects* (B), *selection bias* (C), and *confounding* (D) are also of concern. All biases can occur regardless of directionality between Y and Y^* . Other biases are only applicable in settings matching the orientation of connections shown in this diagram.

quality in human-AI decision-making research, it is critical to assess whether it serves as a satisfactory approximation of the target variable of interest. Measurement theory in the quantitative social sciences provides tools to conduct this assessment by weighing the *construct validity* and *construct reliability* of observed labels [48, 55]. Establishing construct validity and reliability requires evaluating whether an operationalization wholly and consistently represents the latent construct it is purported to measure. This evaluation requires a holistic assessment of numerous converging sources of evidence. Section 4.3.1 details which sources are most salient when examining the validity of target variables in prediction settings.

Measurement error can also be quantified via a *measurement error model*. These models make assumptions on the relationship between the target outcome (Y^*) and its proxy (Y). For example, error models commonly studied in settings with binary target variables include:

- **Uniform:** This error model assumes that the target outcome is randomly corrupted due to exogenous factors unrelated to covariates X , the target outcome Y^* , and model unobservables Z . This setting is also sometimes called *classical measurement error* in statistics and economics, as it assumes that the error term is additive (i.e., $Y^* = Y + \epsilon$) [89]. The machine learning literature more often refers to this setting as *symmetric label noise* [3]. It is possible to learn an unbiased estimate for Y^* when minimizing many common classification performance measures (e.g. misclassification risk) with respect to Y in uniform error settings [73]. Therefore, this error model poses fewer threats to model validity than error processes discussed below.
- **Class-dependent:** This error model assumes that positive ($Y^* = 1$) and negative ($Y^* = 0$) target outcomes are misclassified at different rates. Similarly to the uniform case, measurement error in this setting is uncorrelated with covariates ($Y \perp\!\!\!\perp Y^*|X$) and model unobservables ($Y \perp\!\!\!\perp Y^*|Z$). This model is referred to as *asymmetric* or *class conditional label noise* in machine learning literature [97], and *nondifferential mismeasurement* in statistics and epidemiology literature [82]. In contrast to uniform error settings, training a model to predict a proxy (Y) impacted by class dependent error will lead to inaccurate estimates for the target outcome (Y^*) when minimizing common classification performance measures (e.g., misclassification risk) [73].
- **Feature-dependent:** This model reflects settings in which measurement error occurs *differentially* across sub-populations based on covariates ($Y \not\perp\!\!\!\perp Y^*|X$) or model unobservables ($Y \not\perp\!\!\!\perp Y^*|Z$). Measurement error that varies as a function of covariates is referred to as *differential mismeasurement* in statistics and *feature-dependent label noise* in machine learning literature. This setting is also called *group-dependent error* when the covariate in question is a protected attribute (e.g., gender, race) [105]. Group-dependent error inherits modeling challenges arising in the class dependent case, and has been tied to disparities in criminal justice [81] and medical [1] outcomes in real-world deployments of ADS tools.

Human-AI decision-making research also stands to benefit from existing *methodologies* designed to characterize measurement error in other disciplines. Latent Class Analysis (LCA) is an approach used in psychology and political science to identify latent sub-populations in data that are believed to carry an unobserved characteristic (e.g., personality, political ideology, or disease status) [107]. LCA estimates a set of conditional probabilities mapping multiple discrete *factors* (i.e., *proxies*) to a binary latent variable (e.g., *target outcome*). While LCA is tailored to discrete latent variables, other structural equation models (i.e., factor analysis [50]) are designed for continuous latent variables. Within biostatistics, the Hui-Walter framework is used to estimate the sensitivity and specificity of diagnostic tests in the absence of a gold standard [54]. Given multiple proxies, Hui-Walter can therefore be adapted to estimate the

sensitivity and specificity of each proxy. Like all measurement models, LCA and Hui-Walter make assumptions on the relationship between the target outcome and its proxy. Table 2 states these assumptions in the context of our measurement model taxonomy.

3.4.2 Intervention effects. Evaluations of ADS frequently compare model predictions with human decision-making on the basis of accuracy and statistical fairness measures [46, 102, 108]. However, this practice assumes that decisions and outcomes are statistically independent ($D \perp\!\!\!\perp Y^*$). In contrast, in real-world settings, decisions often serve as *risk mitigating interventions* intended to improve the likelihood of a favorable policy-relevant outcome [7, 25, 60]. In these contexts, past human decisions D_H influence the probability of both the target outcome Y^* and its observed proxy Y (Figure 2.B). Several frameworks have been introduced to study settings in machine learning literature, settings in which decisions impact the are

This setting has been previously cast as a performative prediction problem in machine learning literature [88]. In causal inference, the *treatment effect* of a binary variable (i.e., baseline risk versus a proposed intervention) is defined as its average causal effect on an outcome variable of interest [86].

The role that intervention effects play in target variable bias varies depending on the underlying decision-making setting under consideration:

- **Selective Intervention:** In a *selective intervention* (SI) setting, an organization allocates resources to individuals based on their risk under no intervention. In an education setting, teachers may place students in a tutoring program if they are at-risk of failing the course given no additional resources [100]. In a medical setting, doctors may prescribe costly medical interventions to patients who are at high-risk of stroke given no medical treatment. In child welfare settings, social workers open investigations for cases they believe are at highest risk without agency intervention [57]. In these scenarios, the target outcome Y^* (i.e., students learning the material, biological stroke, and child abuse) is defined under the baseline scenario without intervention as well as under the *treatment* scenario in which an individual receives the intervention under consideration (e.g., tutoring resources, medical care, or social welfare resources).
- **Selective Opportunity:** In a *selective opportunity* (SO) setting, an organization grants individuals an opportunity (e.g., new job, or release from jail pre-trial). In doing so, the organization assesses the risk that individuals will experience an adverse outcome (e.g., poor job performance or recidivism) given that they receive the opportunity under consideration. Typically, individuals are granted the opportunity when the risk under the proposed opportunity is deemed acceptable when balanced with other competing factors [45]. In a selective opportunity setting, the outcome of interest (Y^*) is only relevant among cases that receive the opportunity. For example, decision-makers would not evaluate risk of recidivism under no release on bail. Thus, accounting for intervention effects on the outcome are primarily of concern in a selective intervention setting. However, as we will see in Section 3.4.3, historical decisions also introduce target variable bias in selective opportunity settings via other mechanisms.

In selective intervention settings, a naive comparison of an observational model targeting Y and historical decisions D_H will overlook intervention effects that the intervention induced on the target outcome Y^* and the observed proxy Y . This limitation has been documented in child welfare, where an observational model was shown to *systematically underestimate the risk of welfare cases that benefited most from past child welfare interventions* [25]. In particular, if a social worker elected to open an investigation on a high-risk family, and that investigation averted abuse, the historical data available for modeling will show the case data (X) along with a favorable outcome (i.e., no child removal from home in two years marked by $Y = 0$). In this scenario, the historical decision to intervene was correct despite observational data reflecting no adverse outcome.

The potential outcomes framework developed by the causal inference community offers a useful tool for describing outcomes of interest in selective intervention and selective opportunity settings [96]. Specifically, consider a binary decision setting where $D = 0$ represents the baseline treatment (in SI) or no opportunity (in SO) and $D = 1$ represents the proposed intervention (in SI) or proposed opportunity (in SO). Under potential outcomes, the target outcome of interest Y^* assumes a value of Y_0^* when $D = 0$ and Y_1^* when $D = 1$, while Y assumes a value of Y_0 when $D = 0$ and Y_1 when $D = 1$. This allows us to express the actual outcome observed in historical data in terms of potential outcomes such that $Y = Y_1(D) + Y_0(1 - D)$. The key challenge that arises in this setting is that only one of the two potential outcomes are observed for each case. This is called the fundamental problem of causal inference [86].

In a selective intervention setting, we wish to estimate the risk of the target outcome under no intervention $\mathcal{P}(Y_0^*|X = x)$. In a selective opportunity setting, we wish to estimate the risk of the target outcome under the proposed opportunity $\mathcal{P}(Y_1^*|X = x)$. Previous work by Coston et al. [25] and related work on Individual Treatment Effect (ITE) estimation [98] provides approaches for estimating risk under no intervention using observed proxies. However, intervention effects are particularly challenging when compounded by measurement error. This is because the effect of D on Y^* may differ from the effect of D on Y . For example, it may be possible that an intervention mitigates risk, making an adverse outcome *less* likely, while also increasing monitoring and awareness, making it *more* likely the proxy is observed.

3.4.3 Selection bias. A related challenge introduced by past human decisions D_H is selection bias (Fig 2.D). Selection bias occurs when recorded attributes (X) influenced human decisions (i.e., $X \not\perp\!\!\!\perp D_H$). Because decision-makers may have been more likely to intervene (SI) or grant opportunities (SO) to some sub-populations, this causes some groups to be systematically over- or under- represented in historical outcome data. As a result, ADS models trained on historical data will not perform equally well on all sub-populations during deployment [10]. Selective labels are a strong form of selection bias whereby the outcome proxy is only observed under one historical decision [27, 56, 58, 66]. For example, in child welfare settings, two-year out of home placement is only observed among cases selected for investigation by child welfare workers. This poses a challenge during model development because outcomes are only observed

for those cases accepted by (potentially biased) humans. While selective labels is the most commonly-studied form of selection bias in ADS, this challenge can also arise in decision-making contexts where proxies are observed under both possible decisions (e.g., test outcomes in educational assessment).

Past work has shown that failing to address selection bias can introduce fairness harms [26, 56], perpetuate systematic targeting of historically-disadvantaged groups [6], and lead to under-assessment of the accuracy of past human decisions [66]. This makes it imperative to carefully account for selection bias during development and evaluation of predictive models. While randomizing decisions can mitigate for selection bias, this solution is challenging in high-stakes settings where it is ethically fraught to randomly allocate interventions (SI) or opportunities (SO). However, recent work has made advances addressing selective labels, provided that certain modeling assumptions hold [25, 66, 71, 106].

3.4.4 Confounding bias. An additional challenge arises when *model unobservables* Z are present in decision-dependent outcome tasks. When model unobservables influence both decisions D and downstream outcomes Y^* and Y , this introduces *confounding bias* (Fig 2.D). Confounding bias is problematic in settings where we wish to develop a counterfactual model that predicts an outcome under one or more decisions [86]. Because counterfactual models assume no unobserved confounding (i.e., ignorability), model unobservables can lead to invalid counterfactual model predictions. While approaches have been proposed to address confounding in some settings [24], model unobservables remain a key challenge in decision-dependent outcome settings.

In addition to confounding bias, model unobservables also offer challenges and opportunities for improving ADS [53]. In theory, unobservables provide an opportunity for *complementarity* when human knowledge of contextual information is effectively synthesized with model predictions, which draw on narrower data sources [51, 93]. However, in practice, effectively integrating contextual (Z) and observed (X) information has proven challenging in real-world ADS deployments. Kleinberg et al. [59] show that some contextual information can be predictive of the outcome of interest, while other contextual information can be irrelevant or related to biases in human decision-making. A further challenge arises when decision-makers have limited knowledge regarding which information is available to a model (X) versus unobserved by the model (Z). For instance, in child maltreatment settings, social workers using ADS have reported low trust in model predictions because they believe it does not adequately account for unobserved contextual information. Yet at the same time workers are uncertain which information is or is not available to the model, potentially limiting their ability to calibrate their reliance on the model [57]. As underscored by our findings in section 5, understanding the role of unobservables in ADS remains a key direction for future research.

4 MODEL DEVELOPMENT

We now provide a framework for surfacing target variable modeling assumptions during predictive model development. Growing work in the machine learning community proposes improved modeling approaches for human-AI complementarity (i.e., modifications to the blue region in Figure 1) [52, 69, 102, 108]. These approaches

are developed under the rationale that they predict a target outcome (Y^*) with improved statistical performance measures when compared to past human decisions. However, in practice, models target proxy labels Y observed under the causal diagram described in Figure 1. This limitation is not readily acknowledged by many modeling papers, which mistakenly refer to observed proxies as “ground truth” or a “gold standard” and thus challenges related to target variable bias (Section 3.4).

The development of a predictive model used in ADS involves two distinct modeling steps: *measurement* and *prediction*. During the measurement step, tool developers construct a *measurement model* that operationalizes the target outcome of interest Y^* using observed data available for model development. During the second step, tool designers train a *prediction model* that targets the proxy selected during the measurement modeling step. We now discuss each of these modeling steps in detail.

4.1 Measurement model

During the measurement step, the unobserved outcome of interest (Y^*) is approximated using historical data from the causal diagram in Figure 1. This step involves establishing a *measurement hypothesis* ($\hat{Y}^*$) using observed information: covariates X , past decisions D , and one or more outcome proxies Y . In some settings, a subset of unobservables are also available during model development, but unavailable during deployment. These runtime confounders $Z_r \subseteq Z$ are also present when protected attributes (e.g., race, gender) are available during model development, but are unavailable at deployment time for legal purposes [24, 33]. Given information X , Z_r , D , Y recorded in existing datasets, we can construct a measurement model:

$$\hat{Y}^* = F_m[Y^* | X, Z_r, D, Y] \quad (1)$$

Unlike statistical models commonly used in machine learning contexts, a measurement model cannot be learned from past data because the target outcome Y^* is unobserved. Instead, F_m relies on *measurement assumptions* concerning the relationship between the unobserved outcome of interest and recorded information available for modeling. Therefore, it is not possible to assess the quality of $\hat{Y}^*$ by comparing against held-out data, as is common in prediction settings. Instead, evaluating measurement models requires a multifaceted approach, including assessments of construct validity, synthetic experiments, sensitivity analyses, and other evaluation strategies described in Section 4.3.

All predictive models in ADS introduce a measurement model. However, this model is often *implicitly defined* and makes tacit assumptions on the relationship between available data sources (X , Z_r , D , Y) and the target variable (Y^*). For example, consider the following measurement models proposed in prior academic literature and real-world deployments of ADS:

- One widely-adopted measurement model uses a single observed outcome Y as a proxy for Y^* (i.e., $\hat{Y}^* = \mathbb{1}[Y == 1]$). For example re-arrest is commonly used as a proxy for re-offense in a criminal justice settings [6, 40].
- The first version of the Allegheny County Family Screening tool adopted a combination of two outcome proxies: two-year out of home placement (Y_1) among screened-in cases

Work	Measurement (F_m)	Prediction (F_p)	Assumptions	Bias
Gao et al. [42] Madras et al. [69] Wilder et al. [108] Tan et al. [102] Hilgard et al. [52]	$\hat{Y}^* = F_m[Y]$	Human decisions D_H available at runtime	Proxy and target variables are equivalent $Y^* = Y$	None
De-Arteaga et al. [31]	$\hat{Y}^* = F_m[X, D, Y]$, where $Y^* = D$ for cases in a <i>high consistency set</i> and $Y^* = Y$ otherwise		Assumes Y^* can be identified via high human inter-rater reliability	Measurement error, Selection bias
Lakkaraju et al. [66]	$\hat{Y}^* = F_m[Y]$		Random assignment of instances to humans. Nonuniform acceptance-rates across humans	Selection bias
Coston et al. [25]	$\hat{Y}^* = F_m[Y_d]$, where Y_d is a potential outcome	Human decisions D_H unavailable at runtime	Causal identifiability conditions	Intervention effects
Coston et al. [23]	$\hat{Y}^* = F_m[Y_d, Z_r]$, where Y_d is a potential outcome and Z_r are runtime confounders		Causal identifiability conditions	Intervention effects, Unobservables
Wang et al. [105]	$\hat{Y}^* = F_m[Y]$, where Y is a proxy with group-dependent error		Confident learning assumptions (see [80])	Measurement error
Label noise Menon et al. [73]	$\hat{Y}^* = F_m[Y]$, where Y is observed under mutually contaminated distributions model	Empirical risk minimization with surrogate loss (see [78])	Weak separability assumption for error parameter estimation	Measurement error
Latent Class Analysis McCutcheon [72]	$\hat{Y}^* = F_m[Y]$, where $Y = \{Y^1, \dots, Y^K\}$ are factors related to latent construct	3-step LCA with covariates (see [104])	$Y^i \perp\!\!\!\perp Y^j \mid Y^*$	Measurement error
Hui-Walter Framework Hui and Walter [54]	$\hat{Y}^* = F_m[Y]$, where $Y = \{Y^1, \dots, Y^K\}$ are multiple diagnostic tests	N/A	Test Se/Sp identifiability assumptions	Measurement error

Table 2: Taxonomy of measurement and prediction approaches. Top: methods proposed in ADS literature. Bottom: common methods applied in machine learning, quantitative social sciences, and bio-statistics.

($D = 1$) and two-month re-referral (Y_2) among screened-out cases ($D = 0$) (i.e., $\hat{Y}^* = D \cdot Y_1 + (1 - D) \cdot Y_2$).

- De-Arteaga et al. [31] develops a hybrid measurement model that uses human decisions in cases with inter-expert consistency, and observed proxies otherwise.
- In subjective annotation tasks in which humans directly determine outcome labels, the measurement model aggregates the subjective decision of multiple human annotators $\hat{Y}^* = F_m[D]$, where F_m is a majority vote or more nuanced

procedure (e.g., jury learning [43]). Thus, subjective labels also form *de facto* measurement models.

Table 2 provides a detailed list of the measurement models assumed by existing ADS approaches. This table reifies often-implicit measurement assumptions adopted by prior work. In the bottom three rows, we apply our taxonomy to widely-adopted modeling approaches in machine learning [73, 78], quantitative social sciences [72], and bio-statistics [54] literature.

4.2 Prediction model

After establishing a measurement model to estimate Y^* given (X, Z_r, D, Y) , tool designers then develop a *prediction model* for use in decision-support settings. This prediction model takes observed covariates (X) and predicts the measurement hypothesis ($\hat{Y}^*$) established during the preceding measurement step. Because Z_r and Y are unavailable during deployment, these are not included in the prediction model. Most often, prediction models do not assume human decisions D are available at runtime (i.e., algorithm-in-the-loop [44]). However, in some more nuanced decision-making workflows, models may also assume that human decisions are available at runtime as an additional input (i.e., [42, 69, 102, 108]). Given X and optionally D available at runtime, the prediction model estimates:

$$\hat{Y} = \hat{F}_p[\hat{Y}^* | X, D] \quad (2)$$

Whereas a measurement model is *constructed via measurement assumptions*, the prediction model $\hat{F}_p$ is a *learned mapping* from X (and in some cases D) to the measurement hypothesis $\hat{Y}^*$. Therefore, it is appropriate to evaluate generalization of $\hat{F}_p$ to held-out data via the standard slate of evaluation metrics (e.g., accuracy, AU-ROC, or statistical fairness measures). Critically, this evaluation is conducted with respect to the measurement hypothesis established during the measurement step rather than the target outcome directly. Thus, showing strong performance of $\hat{F}_p$ is *not* sufficient to claim a model generates valid predictions for the target outcome Y^* .

4.3 Measurement model evaluation

Measurement model evaluation introduces a nuanced set of challenges and constraints that may be unfamiliar to machine learning researchers and data scientists. Because the target outcome Y^* is unobserved, it is not possible to directly assess performance with respect to this outcome. Instead, measurement model evaluation requires a holistic, multifaceted approach based on converging sources of evidence. Informed by methods used in statistics, quantitative social sciences, and learning sciences, we provide a recommended set of approaches for validating measurement models in ADS.

4.3.1 Construct reliability and validity. Measurement theory offers a holistic set of criteria for assessing the quality of a target outcome’s operationalization by a measurement model. Under this framework, measurement models are evaluated by weighing evidence for *construct reliability* and *construct validity*.

Construct reliability describes the degree to which a latent phenomena is consistently reflected by a measurement model over time. A key sub-component of reliability is test re-test reliability, which is satisfied in an ADS context if a measurement model generates similar outputs over repeated evaluations. With reference to the ADS measurement model provided in e.q. 1, this means that the same set of inputs (X, Z_r, D, Y) should map to the same measurement hypotheses $\hat{Y}^*$ over repeated evaluations. This criterion is likely to be satisfied in settings where target variables are assigned via well-established organizational protocols (e.g., medical outcomes). However, test re-test reliability is a greater concern in settings such as toxicity or hate-speech detection, where target variables are

assigned via subjective human judgement and where the operationalization of the target construct can vary substantially based on rater identity [32, 32], context [84], and variability in annotation protocols [90].

Construct validity describes the extent to which a measurement model adequately captures an unobserved phenomenon of interest. In predictive modeling settings, construct validity of target outcomes assesses the degree to which outcome proxies (e.g., 2-year placement in foster care, re-arrest) wholly and completely reflects the broader target phenomena being represented in a prediction model (e.g., child abuse and neglect, risk to society, or creditworthiness). A measurement model satisfies construct validity if F_m (e.q. 1) produces measurement hypotheses $\hat{Y}^*$ that is consistent with the target variable of interest Y^* . Thus, while construct reliability is roughly analogous to the notion of statistical variance in F_m , construct validity is analogous to statistical *bias* in F_m . Jacobs and Wallach [55] provides an overview of measurement-related concerns as they relate to algorithmic fairness, while Coston et al. [22] provides an overview of sub-components of validity salient to assessing the validity of RAI models.

4.3.2 Outcome cross-validation. In many settings, multiple outcome proxies are available that are believed to be related to the target outcome of interest. For instance, in criminal justice settings, courts often track outcomes including failure to appear, 2-year general recidivism, and 2-year violent recidivism [40]. In child welfare settings, government agencies may track outcomes such as substantiation of abuse allegations, acceptance for welfare services, agency re-referral, placement in foster care, and hospitalization [103]. These reference outcomes can be leveraged to assess the quality a measurement model adopted by an ADS tool.

One strategy for improving measurement model evaluation using reference outcomes is *outcome cross-validation*. In this approach, a model is trained to predict one outcome proxy Y^1 , and is then evaluated on a slate of additional held-out variables $\{Y^2, \dots, Y^K\}$ that modelers expect may be reasonable proxies for the outcome of interest. If targeting Y^1 results in strong prediction performance across held-out proxies, this provides evidence that adopting the proxy under consideration serves as a suitable measurement model. The term “*outcome cross-validation*” was introduced within the educational assessment community to validate metrics for latent constructs such as student motivation [92]. Kleinberg et al. [58] independently developed the same procedure to evaluate pre-trial risk assessments using a slate of crime-related outcomes, in the presence of omitted payoffs (see Table V of [58]). Most recently, De-Arteaga et al. [31] use outcome cross-validation to assess a proposed modeling approach using held-out reference outcomes not targeted during model training.

Special cases of outcome cross-validation are often used to assess sub-components of construct validity. For example, a model demonstrates *predictive validity* if its predictions correlate with a held out future outcome that is known to be related to the target phenomena of interest. Similarly, a model demonstrates *convergent validity* if it generates predictions that are correlated with a held out outcome that is already known to satisfy construct validity. Conversely, if a model’s predictions are *not* correlated with a held

out outcome unrelated to the outcome of interest, this provides evidence for *discriminant validity*.

For example, developers of the Allegheny County Family Screening Tool assessed predictive validity by evaluating whether the tool – which was trained to predict a 2-year out of home placement proxy – generates predictions that are correlated with future child abuse related hospitalization outcomes. The developers similarly assess discriminant validity by showing that model predictions are *not correlated* with non-abuse related child health outcomes (e.g., cancer).

4.3.3 Sensitivity analyses. Sensitivity analyses offer an additional tool for characterizing the vulnerability of proxies to unforeseen target variable bias. These analyses answer the question: “*what magnitude of unobserved bias is required to nullify observed findings?*”. Sensitivity analyses have been used in causal inference settings to estimate the magnitude of unobserved confounding (i.e., unobservables Z) necessary to invalidate a causal effect estimate [34, 95]. This approach has also been applied to predictive modeling. More recently, sensitivity analyses have been developed to assess the degree of measurement error necessary to nullify findings from statistical fairness assessments conducted via proxy labels [38].

Sensitivity analyses enable assessing the degree of *measurement model misspecification* permissible before evaluation of a *prediction model* is invalidated. Therefore, evaluations of predictive models tools can report sensitivity analysis results to provide further evidence that results are robust to a given degree of unobserved confounding, selection bias, and measurement error. Future evaluations would also benefit from sensitivity analysis frameworks that enable jointly considering several (potentially additive) sources of bias impacting outcome variables in parallel.

4.3.4 Synthetic evaluation. The approaches introduced above assess measurement models using observational data. A key benefit of these assessments is that they can be used to evaluate measurement models on realistic datasets that will be encountered during model deployment. However, a core limitation of these approaches is that one never knows the true relationship between the outcome of interest Y^* and observed proxies Y in naturalistic data. This poses a challenge for conducting rigorous assessments of proposed modeling approaches. Instead, model-level evaluations in ADS typically circumvent this issue via *synthetic evaluations* that test whether proposed approaches are robust to experimentally manipulated sources of statistical bias [25, 31, 73, 105].

An additional limitation of synthetic evaluations is that they require assuming a specific measurement error model. If the data generating process adopted in a synthetic evaluation does not adequately reflect realistic conditions, this can lead to a false conclusion that the approach is robust under real-world conditions. This concern is salient because to date, synthetic evaluations are often designed with bespoke data generating processes intended to highlight the specific source of target variable bias being investigated by the work (e.g., selection bias, measurement error).

4.3.5 The oracle test. Chouldechova et al. [21] propose a thought experiment called the “the oracle test”, which offers a conceptual tool for probing unforeseen challenges arising from target variable bias. This thought experiment supposes that we have access to an

oracle model that can predict an observed proxy Y with perfect accuracy. The key question posed by this test is: “*What concerns remain given access to an oracle?*”. Because we have a “perfect” prediction model (i.e., e.q. 2), remaining concerns are often related to measurement and validity. By applying the oracle test to a child welfare maltreatment decision-making setting, Chouldechova et al. [21] surface several additional issues, including measurement error. Furthermore, Green and Chen [45] adopt an argument akin to the oracle test by arguing that improvements to predictive accuracy do not equate to improved policy outcomes when competing factors in addition to risk (i.e., defendant liberty) play a role in judicial decision-making.

5 ASSESSING GAPS AND OPPORTUNITIES FOR EXPERIMENTAL RESEARCH

Our causal diagram can be used to assess the extent to which existing in vitro studies consider measurement error, intervention effects, and related challenges (Section 5.1). By mapping existing studies to corresponding sub-regions of our proposed diagram, we surface key blind-spots in our current understanding of human-AI decision-making. Our causal diagram can be used to evaluate whether experimental findings are likely to hold *ecological validity* in a given real-world ADS deployment (Section 5.2). More broadly, the methodology of causal diagrams can also serve as a scaffolding for structuring empirical knowledge beyond concerns related to target variable bias (Section 5.3).

5.1 Mapping existing experimental study designs to our causal diagram

To assess the extent to which existing experimental studies examine factors related to target variable bias in their study design, we revisit a comprehensive literature review conducted by Lai et al. [63] through the lens of our causal diagram (Figure 1). Lai et al. [63] review over one hundred experimental studies of algorithm-assisted decision-making published in premiere venues between 2018 and 2021. Our follow-up analysis extends this review to studies published in 2022 at the same set of venues, in addition to other recently published pre-prints. We further limit selection criteria applied by Lai et al. [63] to studies examining human-AI decision-making in prediction-based decision-making settings (i.e., scope outlined in Section 3.1). Thus, we exclude studies included in the initial review with a focus on NLP-related tasks (e.g., Text QA, text summarization).

We find that 66 out of 72 ($\approx 92\%$) studies satisfying our criteria conduct experimental evaluations focusing on a narrow sub-graph of our causal diagram. These studies most frequently investigate a modification to the joint human-algorithm decision-making process (i.e., the blue D_H and D_A region) using observed attributes X and an outcome proxy Y (Figure 3). Such studies assume that (1) the target variable and proxy are equivalent (i.e., limited discussion of measurement error or validity concerns), (2) all predictive attributes of interest are observed by both the algorithm and the human (i.e., limited consideration of model unobservables), and (3) decisions and outcomes are statistically independent (i.e., limited consideration of intervention effects).

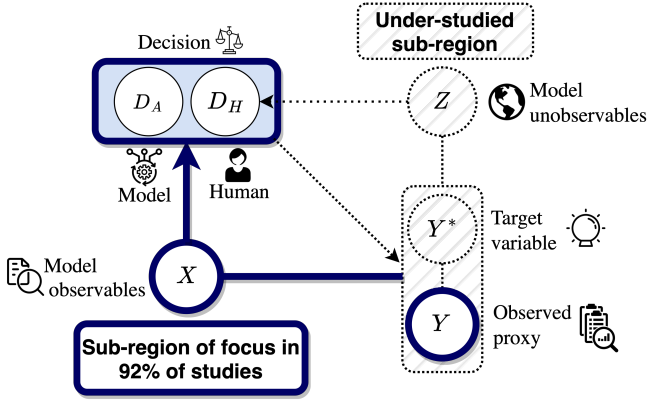


Figure 3: 66 of the 72 studies ($\approx 92\%$) in our review examine a narrow sub-region of our proposed causal diagram.

Six of the remaining studies we review examine different sub-regions of the causal diagram described in Figure 1. Table 3 groups these studies by the sub-region under study, including unobservables [51, 53], measurement error [43], selection bias [87], and omitted payoffs [39, 45]. While this work offers an early lens into how target variable bias can impact algorithm-assisted human decision-making, our empirical understanding of these challenges remains limited compared to the joint human-algorithm decision region investigated by $\approx 92\%$ of studies. For instance, no studies in our review examine the role of intervention effects in algorithm-assisted decision-making settings. Additionally, no studies examine *interacting sources of bias*, such as measurement error and unobservables. Given the prevalence of compounding challenges in real-world settings (e.g., Table 1), this presents a pressing gap in our current empirical understanding of algorithm-assisted *human* decision-making, and opens a broad space of open questions and future research opportunities.

5.2 Assessing ecological validity of in vitro studies

The gap we identify between in vivo challenges (i.e., Table 1) and in vitro studies (i.e., Figure 3) carries implications for the ecological validity of experimental studies. Threats to ecological validity may be most acute when findings from a controlled study conducted under simplified conditions are *generalized* to real-world ADS deployments in which multiple sources of target variable bias are present. In these settings, measurement error and intervention effects could impact whether findings gathered via controlled experiments also apply in more complex real-world conditions.

Fortunately, our causal diagram provides a tool for assessing whether findings from an *in vitro* study are likely to generalize to a given *in vivo* ADS tool deployment. The first step in this process involves mapping the decision support task to its corresponding DAG (e.g., identifying whether the task involves decision-dependent vs. decision-independent outcomes). Next, based on domain expertise, one can identify whether different sources of bias are likely to be relevant in the given real-world deployment. For instance, a model deployed to allocate tutoring resources (selective intervention) may

Work	Setting	Sub-region of causal diagram
Hemmer et al. [51] Holstein et al. [53]	House price prediction	Unobservables: $D \leftarrow Z \rightarrow Y$
Gordon et al. [43]	Toxicity detection	Measurement error: $Y^* \rightarrow Y$
Peng et al. [87]	Hiring	Selection bias: $X \rightarrow D$
Green and Chen [45] Fogliato et al. [39]	Judicial	Omitted payoffs: $Q \rightarrow D_H$

Table 3: Experimental studies examining the under-studied sub-region provided in Figure 1

need to account for measurement error in learning outcomes and intervention effects from historical tutoring decisions. In contrast, a model deployed for a perceptual assessment task (e.g., predicting current forest cover from limited image data) may not need to address these concerns. After identifying the DAG and relevant sources of bias, one can assess whether an experimental study is likely to generalize to this setting by examining whether the study was also conducted under similar structural conditions (i.e., also tested decision-dependent or decision-independent outcomes).

To demonstrate how causal diagrams can be used to assess ecological validity of lab-based studies, consider a previous lab-based assessment conducted by Park et al. [83]. This study – which is sampled from the 66 studies covered by the blue sub-region of the causal diagram provided in Figure 3 – examines whether introducing a delay between when humans view observed features X and algorithmic recommendations D_H improves their performance on a perceptual jelly bean counting task. Because the true quantity of jelly beans does not depend on the decision under consideration, this setting involves decision-independent outcomes. Further, the influence of human-only observed attributes Z and measurement error is limited in this task. Therefore, findings from this work may most readily generalize to real-world decision-making settings with limited interference from measurement error, model unobservables, and treatment effects.

5.3 Scaffolding a science of human-AI decision-making

In this work, we develop a causal diagram that characterizes sources of bias impacting target variables. However, beyond this focus, causal diagrams also offer a powerful scaffolding for studying other aspects of human-AI decision-making. While we model the target variable (Y^*) endogenously as a function of (X, Z, D, Y) , one could also construct a causal diagram examining the joint human-algorithm decision-making process. For example, Green and Chen [45] specify such a DAG that models how humans weigh risk against other competing factors (e.g., culpability, value of defendant freedom) during pre-trial release decisions. The authors then experimentally verify a *hypothesized* edge in this DAG via a controlled

online study. Through a series of such studies, it may be possible to develop a more comprehensive, grounded understanding of algorithm-assisted human decision-making across decision-support tasks. This process of specifying, testing, and refining causal models is central to existing empirical disciplines, including cognitive science and sociology [85].

6 DISCUSSION

Our work surfaces a disconnect between the challenges that arise in *real-world deployments* of algorithmic systems versus *current research practices* adopted in the literature on human-AI decision-making (i.e., experimental study designs, modeling assumptions, measures of human-AI decision quality). Addressing this disconnect is critical in light of the grave real-world impacts introduced by target variable bias, including misallocation of medical resources [81] and perpetuation of historical bias in the criminal justice system [1, 6, 40], among many others (Table 1).

Our work provides a critical first step for addressing this disconnect by clarifying the relationship between measurement error, intervention effects, unobserved confounding, and selection bias via intuitive causal diagrams. Going forward, we hope that this framework will support more comprehensive assessment of modeling assumptions (Sections 4.1 and 4.2), rigorous evaluation of measurement-models (Section 4.3), and experimental human subjects studies that investigate the implications of bias in algorithm-assisted *human* decision-making (Section 5). However, our work is only the first step towards comprehensively addressing target variable bias in human-AI decision-making research.

In particular, future research should develop holistic measures of decision-quality that reflect factors beyond statistical performance computed via a single outcome proxy. These measures should reflect both *process-oriented* considerations (i.e., how multiple decision-relevant factors are weighted [44], and adherence to procedural, interpersonal, and informational justice) in addition to *outcome-oriented* considerations (i.e., whether a decision led to a beneficial outcome). Where possible, outcome-related measures should draw upon *multiple decision-relevant proxies* to better-account for limitations of adopting any single proxy in isolation. While this practice is standard in disciplines such as learning sciences, diagnostic medical testing, and psychology, to date, human-AI decision-making research has primarily adopted outcome-oriented measures that hinge upon on a single potentially flawed proxy.

Our work also motivates exciting new lines of human-AI decision-making research. For instance, our review of experimental human subjects research (Section 5.1) identifies an open space of rich empirical questions to be investigated through future studies. We currently have an incomplete understanding of how measurement error impacts human trust and reliance on algorithmic predictions. Further, while counterfactual prediction methods have been proposed in the modeling literature, we currently have limited knowledge of how these methods can be leveraged to support more effective human decision-making practices. More generally, our causal framework provides a set of tools for (i) identifying open empirical questions, (ii) designing studies with robust ecological validity, and (iii) synthesizing findings from multiple experimental studies into

a more complete scientific understanding of human-AI decision-making.

We hope that our work will raise awareness of target variable bias in the human-AI decision-making research community and spur efforts to better align research practices with the complex challenges encountered in real-world deployments of algorithmic systems.

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